KHARAGPUR **DATA ANALYTICS** GROUP

SKIN CANCER DETECTION

TEAM 4

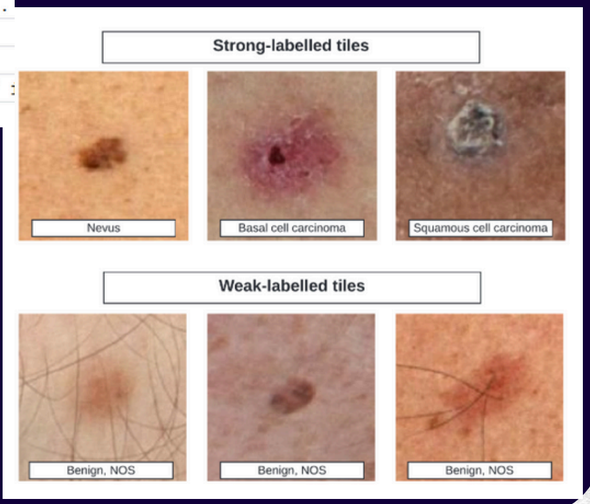
PROBLEM STATEMENT

OVERVIEW

To build a highly accurate malignant cancer cell detection deep learning model trained on a massive dataset of over 4,00,000 entries, including their corresponding high-resolution images and vital subject information such as age and gender. The model will use this intel to differentiate friend from foe.

METADATA

field name	description
target	Binary class {0: benign, 1: malignant}
lesion_id	Unique lesion identifier. Present in le
iddx_full	Fully classified lesion diagnosis.
iddx_1	First level lesion diagnosis.
iddx_2	Second level lesion diagnosis.
iddx_3	Third level lesion diagnosis.
iddx_4	Fourth level lesion diagnosis.
iddx_5	Fifth level lesion diagnosis.
mel_mitotic_index	Mitotic index of invasive malignant mel
mel_thick_mm	Thickness in depth of melanoma invasion
tbp_lv_dnn_lesion_	Lesion confidence score (0-100 scale).
isic_id	Unique case identifier.
patient_id	Unique patient identifier.
age_approx	Approximate age of patient at time of :
sex	Sex of the person.



DATASET EXPLAINED

• GIVEN DATA EXPLAINED

The provided dataset consisted of three key components: (1) subject data — a 401,059 × 55 dataframe containing 55 features, (2) a metadata file describing each of these columns, and (3) an image file in HDF5 format containing high-resolution images of various skin lesion areas.

Target Columns

- **Target columns are a collection of columns that contain information regarding it**
- **These contain target, iddx_1 to iddx_5, iddx_full, mel_mitotic_index, mel_thick_mm**
- **The iddx columns are hierarchical diagnostic labels for each lesion. e.g. iddx_1 = broad first-level class, iddx_2 = more specific second level, iddx_3 = finer subclass,iddx_4 and iddx_5 are just deeper (4th- and 5th-level) diagnostic labels and iddx_full = the full/complete diagnosis string**

Patient Columns

- **These columns consist of features that describe the physical characteristics of the patient, namely age_approx and sex. They have correlations of 0.011 and 0.002, respectively, with the target**
- **Null Values of age_approx are taken as the median of the age_approx**
- **Null Values of sex are taken as the mode of the sex**

DATASET EXPLAINED

Tbp_basic Columns

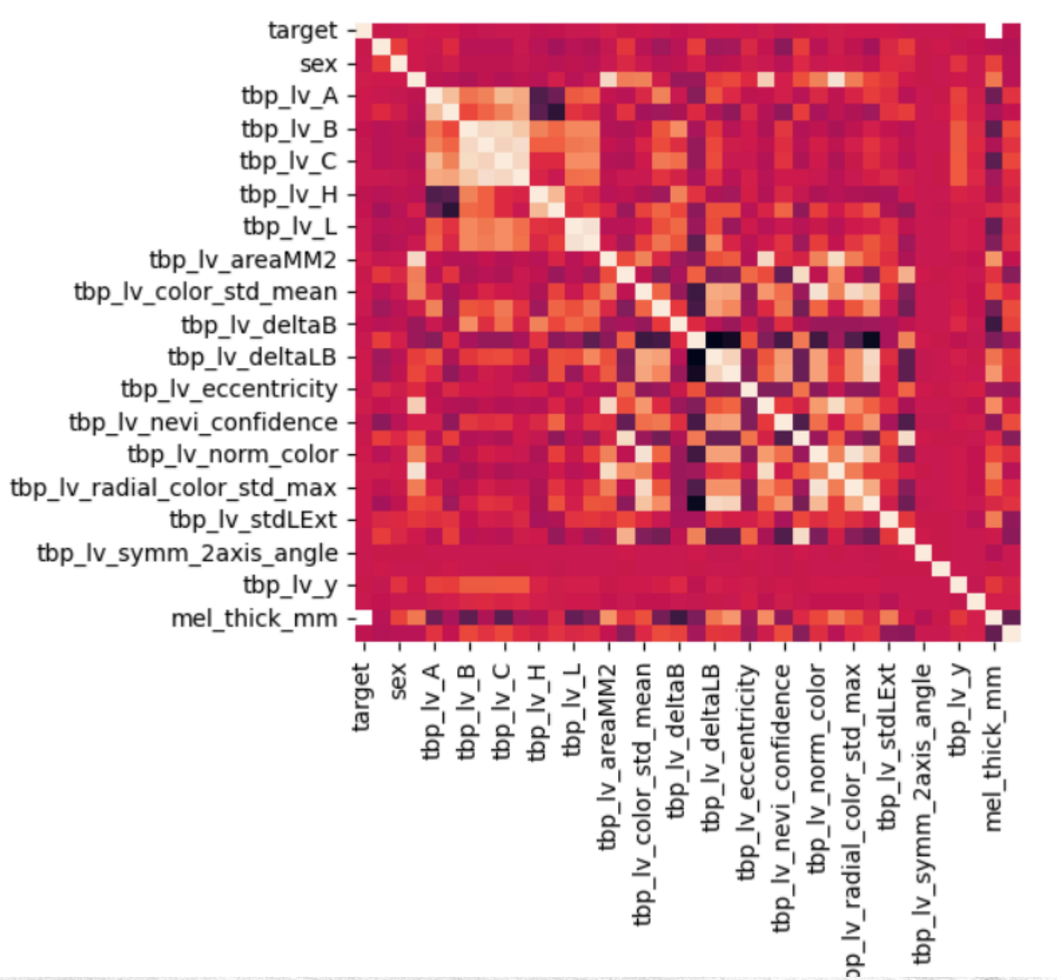
- Consists of several columns including isic_id, tbp_lv_areaMM2, tbp_lv_area_perim_ratio,etc.
- The tbp_lv_ are image-derived lesion-level features (computed from the 3D Total Body Photography). They describe size, shape, color, contrast and some classifier/confidence scores for each lesion tile
- In this features like tbp_lv_areaMM2, tbp_lv_perimeterMM, and clin_size_long_diam_mm are the features that are correlated with target. They reprsent the dimensions of the lesion on the skin.
- Features like tbp_lv_color_std_mean, tbp_lv_radial_color_std_max — measures of color irregularity / asymmetry inside the lesion

Tbp_extra Columns

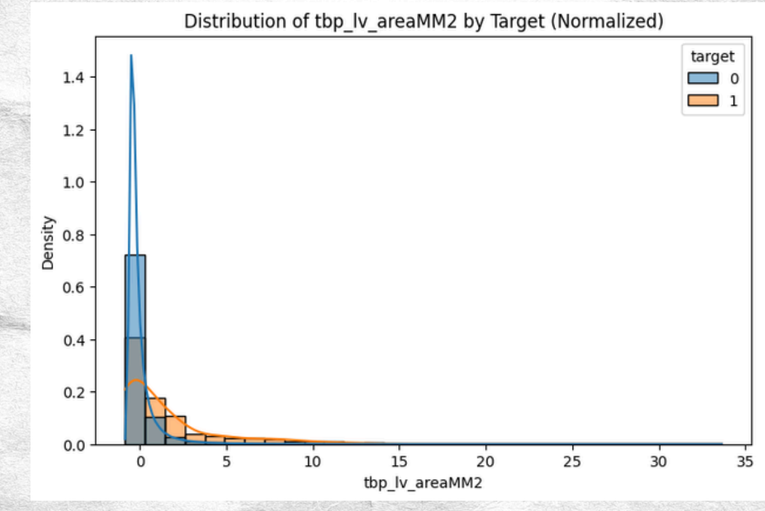
- Consists of several columns including tbp_lv_A, tbp_lv_Aext, tbp_lv_B, etc.
- The tbp_lv_x represents the X-coordinate of the lesion on 3D Total Body Photography. Similarly, the other columns describe additional numerical features analyzed about the physical characteristics of the lesion, which might be important in our analysis
- Features like tbp_lv_deltaA and tbp_lv_deltaB represent the average contrast of inside vs. outside lesion and tbp_lv_stdLExt represent the standard deviation of L outside lesion.
- Features such as tbp_lv_dnn_lesion_confidence, tbp_lv_deltaLBnorm and tbp_lv_B have good correlation with the target

EXPLORATORY DATA ANALYSIS

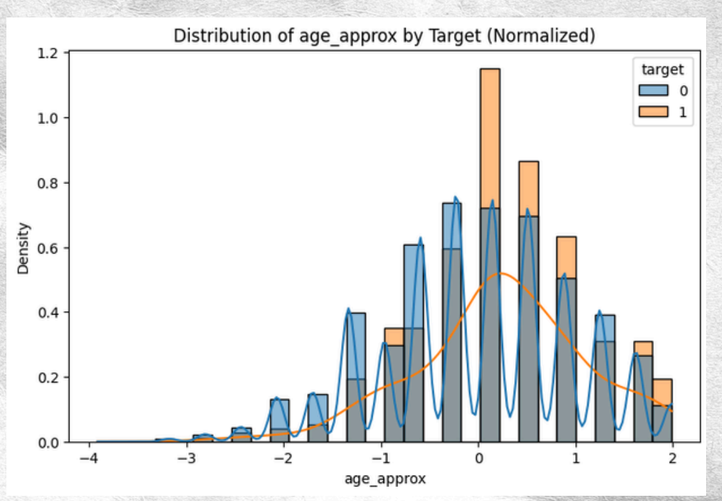
- **Correlation Heatmap**



- **tbp_lv_areaMM2 density**

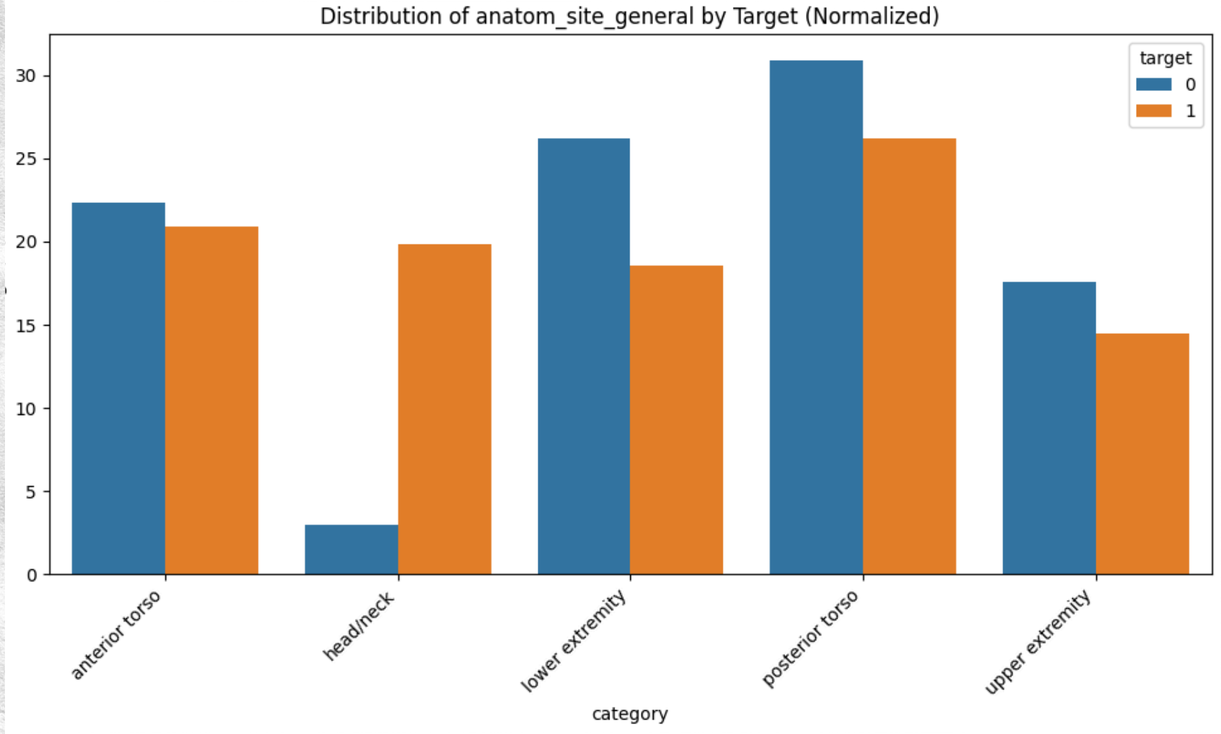


- **Age Distribution**

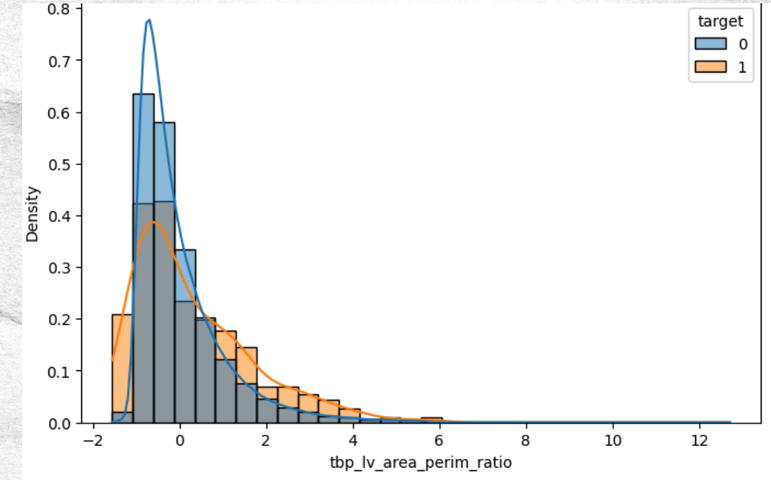


EXPLORATORY DATA ANALYSIS

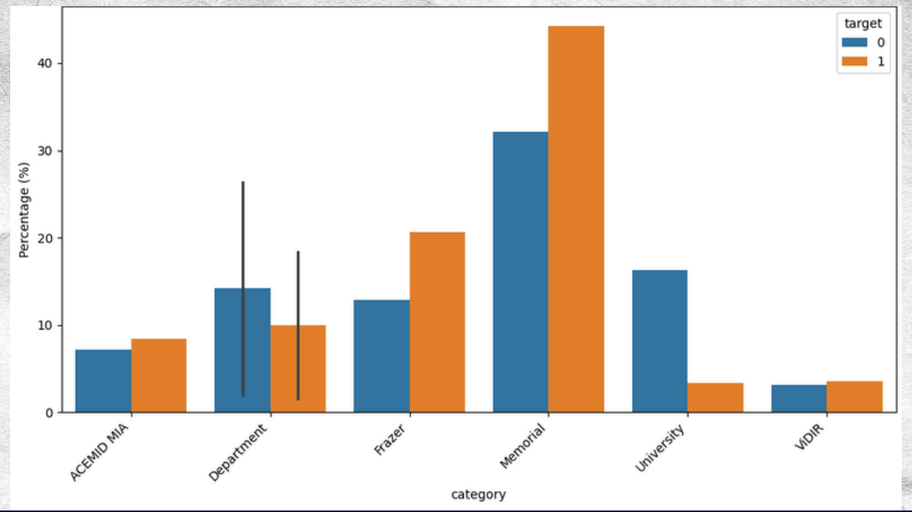
- **Anatom site general by Target**



- **tbp_lv_area vs perimeter ratio**

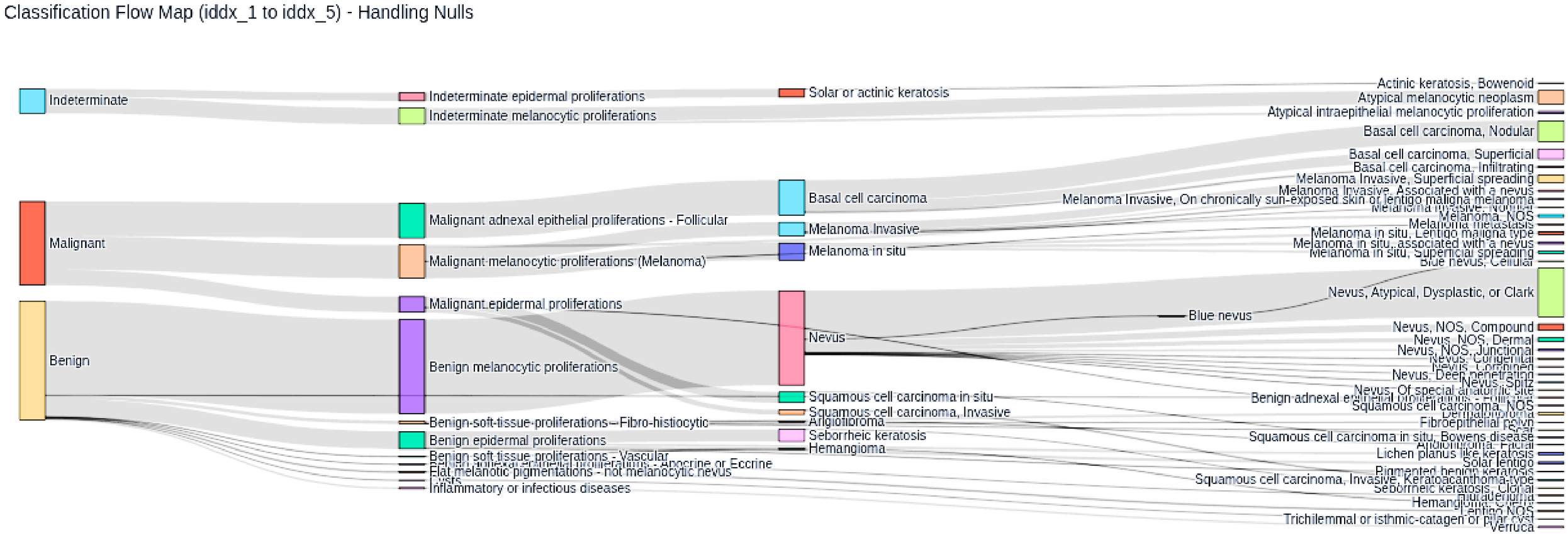


- **Place of detection distribution**



EXPLORATORY DATA ANALYSIS

- **Detailed category graph for malignant**



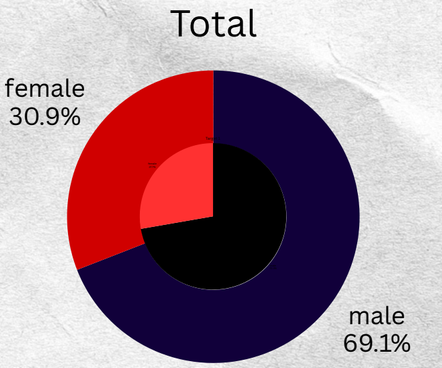
APPROACH EXPLAINED

• STEP 1- DATA PREPROCESSING

Before training, the raw data needed to be cleaned and transformed.

Handling missing values and Imputation

- **Missing sex data was filled with the most common value ("male"), ensuring every entry had a gender label.**
- **Missing age_approx data was filled with the median age of the dataset, providing a reasonable central value.**
- **Missing anatom_site_general values were randomly assigned from the three most common body sites to preserve the overall distribution.**



Feature Slection

- **We identified and selected the most relevant features for our model, including patient demographics and various lesion characteristics, while excluding irrelevant columns.**

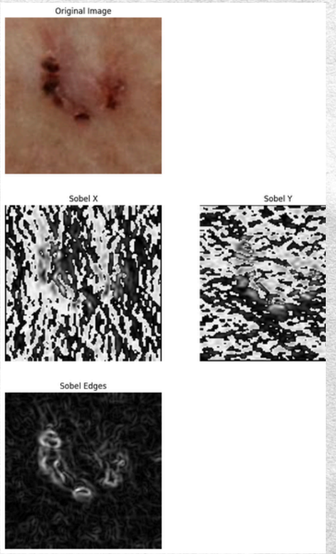
APPROACH EXPLAINED

• STEP 2 - DATA ENCODING AND SCALING

Making the Data Understandable for the Model.

Categorical Encoding

- **Many features, like sex, anatom_site_general, and attribution, are categorical (non-numerical). We converted these into a numerical format using a process called One-Hot Encoding, which creates new binary columns for each category. This allows the model to interpret these features without implying any incorrect order.**



Numerical Encoding

- **We used StandardScaler on our numerical features (like age and lesion measurements). This is a crucial step that standardizes the data by removing the mean and scaling to unit variance. This prevents features with larger values from disproportionately influencing the model's training process.**

APPROACH EXPLAINED

• STEP 3 - ADDRESSING DATA IMBALANCE

Balancing the Dataset for a Fairer Mode - Resampling Strategy

The initial dataset was highly imbalanced, with a very small percentage of malignant cases (target = 1)

Training a model on imbalanced data can lead to poor performance, as it may simply learn to predict the majority class.

To address this, we employed a resampling strategy to adjust the dataset's composition for our analysis.

- **The smaller, "minority" class (malignant cases) was oversampled to create more data points.**
- **The larger, "majority" class (benign cases) was undersampled by taking a random selection of 10,000 entries.**
- **We applied resampling only to the training data to prevent leakage and ensure realistic model evaluation.**

APPROACH EXPLAINED

STEP 4 - IMAGE PREPROCESSING

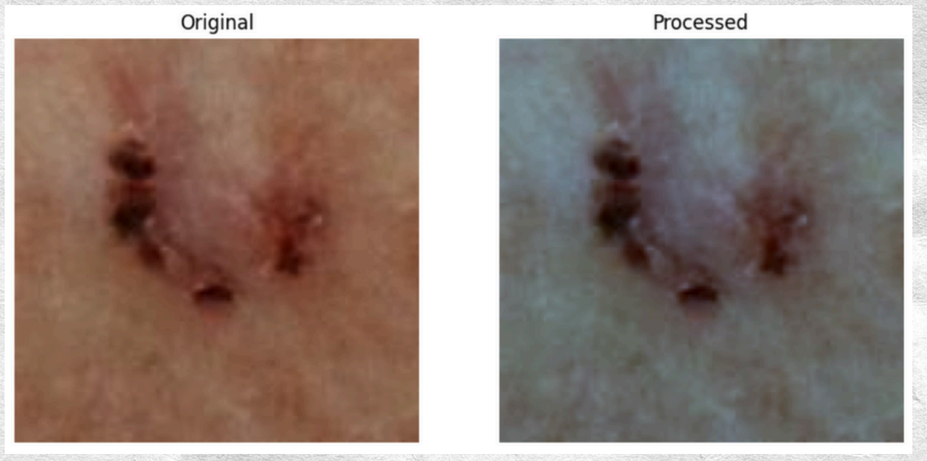
Preprocessing the image for feature extraction

Techniques Attempted

- **Initial Approach focused heavily on enhancement of contrast of the image, removal of noise in the form of hair and lesion extraction.**
- **However, upon testing , hair removal and lesion extraction , were abandoned as it lead to decrease in precision and recall.**

Results

Applying Color Constancy



Lesion Extraction

APPROACH EXPLAINED

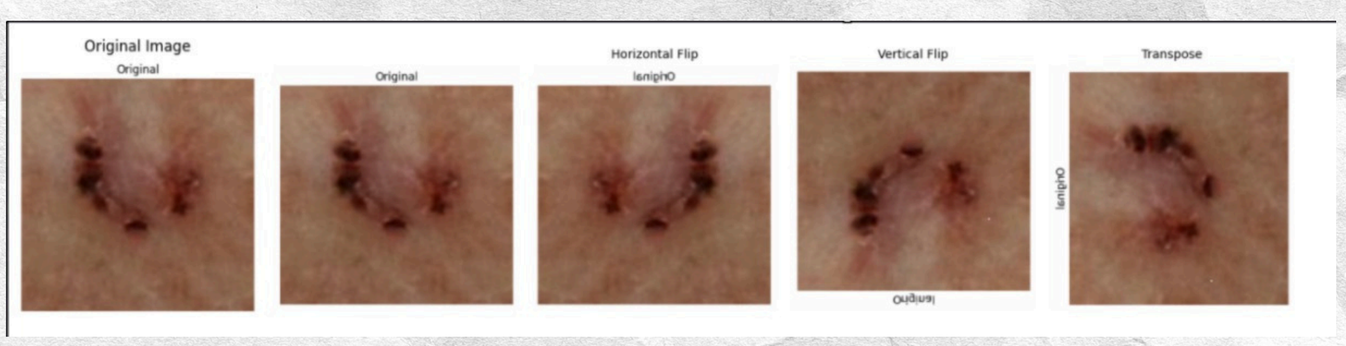
• STEP 4 - IMAGE PREPROCESSING

Preprocessing the image for feature extraction

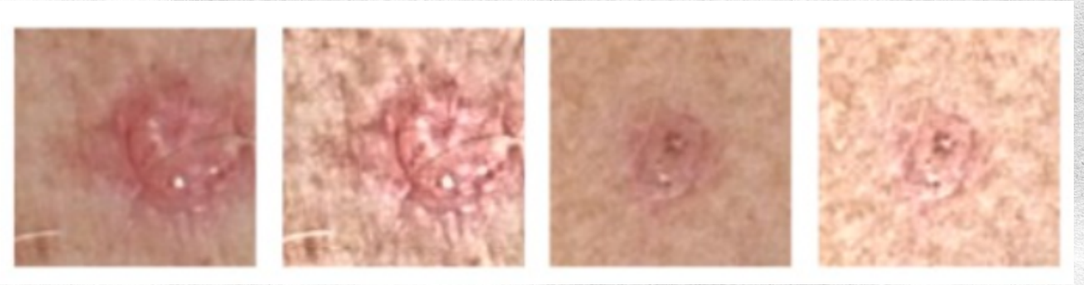
• Techniques Used

- **The dataset was preprocessed to address class imbalance and improve lesion visibility. Oversampling of minority classes was performed using augmentation techniques such as zoom-in, zoom-out, rotation, and flipping, ensuring better diversity in the training data.**
- **Contrast enhancement was applied to highlight and enhance the lesion regions of the skin, making important features more distinguishable for the models.**

• Results



Augmentation



Contrast Enhancement

OTHER MODELS

• STEP 4 - TRYING DIFFERENT MODELS

Trying logistic regresssion model on the dataset

• MOTIVE

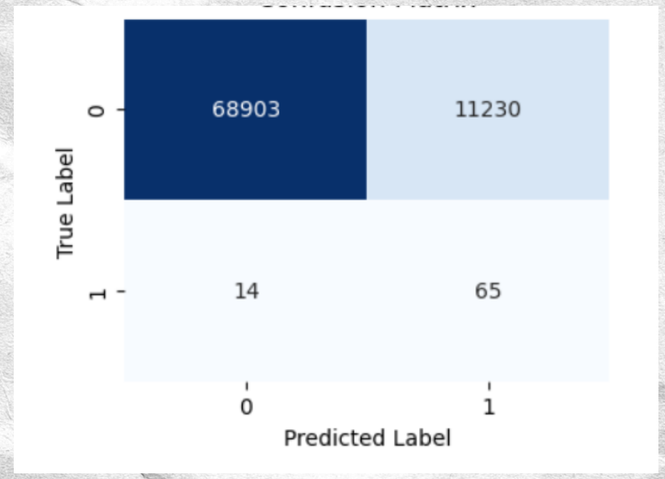
We use logistic regression as a principled, low-complexity baseline that yields well-calibrated probabilities and interpretable feature weights.

• RESULTS

• Classification Report

	precision	recall	f1-score	support
0	1.00	0.86	0.92	80133
1	0.01	0.82	0.01	79
accuracy			0.86	80212
macro avg	0.50	0.84	0.47	80212
weighted avg	1.00	0.86	0.92	80212

• Confusion Matrix



OTHER MODELS

• STEP 4 - TRYING DIFFERENT MODELS

Trying XGBoost model on the dataset

• MOTIVE

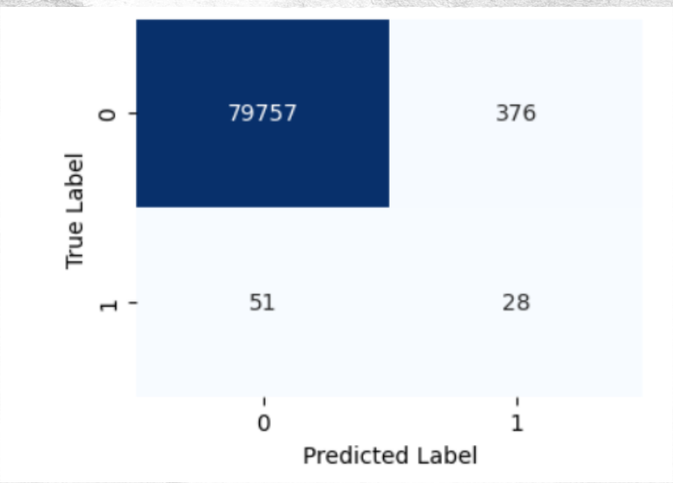
We use XGBoost as a robust nonlinear baseline that captures feature interactions missed by linear models while remaining fast, regularized, and interpretable via feature-importance and SHAP analyses.

• RESULTS

• Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	80133
1	0.07	0.35	0.12	79
accuracy			0.99	80212
macro avg	0.53	0.67	0.56	80212
weighted avg	1.00	0.99	1.00	80212

• Confusion Matrix



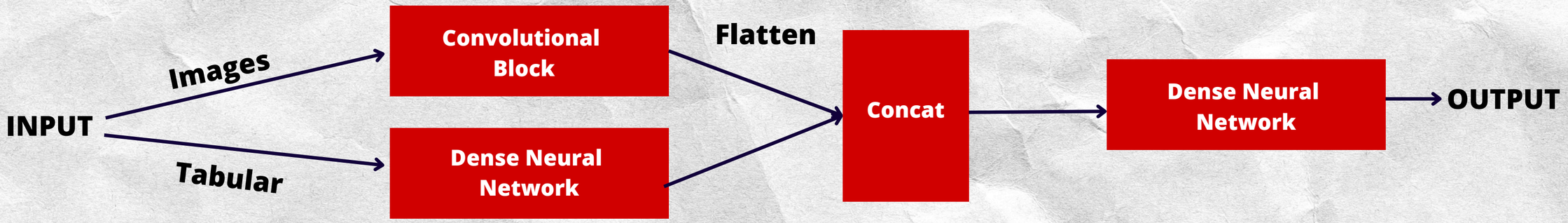
MAIN MODEL

• STEP 5 - DESIGNING THE MAIN MODEL

Trying ResNet, EffNet and ViT to make the final model

• RESNET 50

ResNet is leveraged for its residual connections that facilitate deeper networks to effectively learn hierarchical features, allowing the model to overlook noise and capture meaningful patterns even from sparse minority class samples due to its depth.



• EffNet

Chosen for its optimized balance of network depth, width, and resolution scaling, which enhances overall performance across multiple metrics by efficiently utilizing model capacity and computational resources.

• Vision Transformer

Incorporated to explore whether important spatial and localized patterns exist, especially relevant for detecting skin lesion cancers where critical features tend to be localized rather than globally distributed.

MAIN MODEL

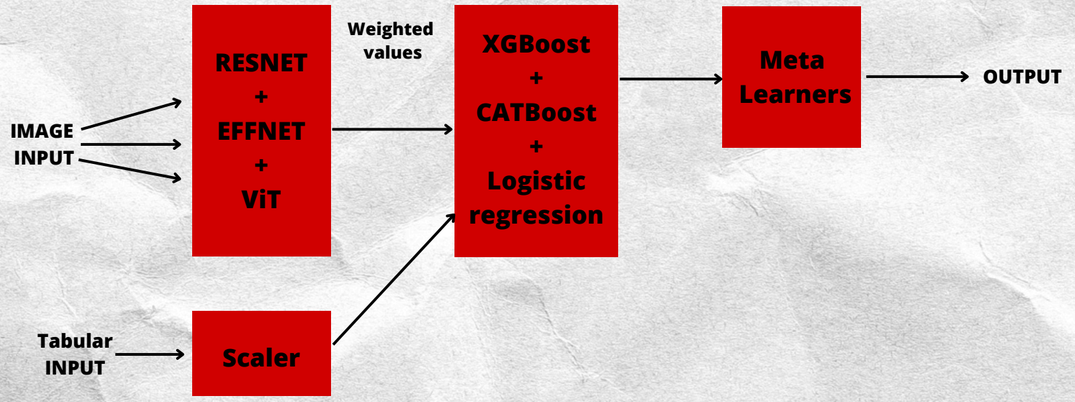
• STEP 5 - DESIGNING THE MAIN MODEL

Ensembling all three to get the best possible results

• Ensembled Model

The architecture is a multi-modal stacked ensemble designed to combine image and tabular data for skin cancer prediction. The image input is processed separately through three powerful feature extractors—ResNet, EfficientNet, and Vision Transformer (ViT)—while the tabular input is normalized using a scaler. Their outputs are assigned learned weights and fused into a joint representation, which is then fed into multiple learners: XGBoost, CatBoost, and Logistic Regression. Finally, a meta-learner combines these model predictions to generate the final output, leveraging the complementary strengths of CNNs, transformers, and boosting methods for more accurate and robust classification.

• Architecture



• Results and comparison

```
threshold = 0.3571428571428571
roc_auc_score = 0.8308466122828523
              precision    recall  f1-score   support

           0       0.97      0.86      0.91       802
           1       0.45      0.81      0.58       118
```

Metrics of single resnet model

```
Final ROC AUC: 0.9094530622596052
              precision    recall  f1-score   support

         0.0     0.9663    0.8217    0.8881       802
         1.0     0.3992    0.8051    0.5337       118
```

Metrics of ensembled model

CONCLUSION

• Some conlusions from the model

- Resnet gives best recal
- resnet gives best recall on minority class, better than efficientNet beacuse ResNet's deep residual architecture allows it to learn detailed features from scarce minority class samples, boosting recall, while EfficientNet's balanced scaling of network depth and width prioritizes overall accuracy and efficiency, which can limit its focus on rare class patterns.
- In medical contexts, depending on the clinical use case, improving overall discrimination (ROC-AUC) can be very valuable even if precision is roughly the same, especially if recall (sensitivity) is better, which means more true cases are detected.

• Comparison of results

threshold = 0.26530612244897955
roc_auc_score = 0.8137389576905196

	precision	recall	f1-score	support
0	0.98	0.72	0.83	802
1	0.32	0.91	0.48	118
accuracy			0.74	920
macro avg	0.65	0.81	0.65	920
weighted avg	0.90	0.74	0.79	920

Metrics of single resnet model (best recall)

threshold = 0.3571428571428571
roc_auc_score = 0.8308466122828523

	precision	recall	f1-score	support
0	0.97	0.86	0.91	802
1	0.45	0.81	0.58	118
accuracy			0.85	920
macro avg	0.71	0.83	0.74	920
weighted avg	0.90	0.85	0.87	920

Best auc roc score of resnet

Final ROC AUC: 0.9094530622596052

	precision	recall	f1-score	support
0.0	0.9663	0.8217	0.8881	802
1.0	0.3992	0.8051	0.5337	118
accuracy			0.8196	920
macro avg	0.6827	0.8134	0.7109	920
weighted avg	0.8935	0.8196	0.8427	920

Metrics of ensembled model

Muti-Stage Ensemble

The following is the architecture of this model

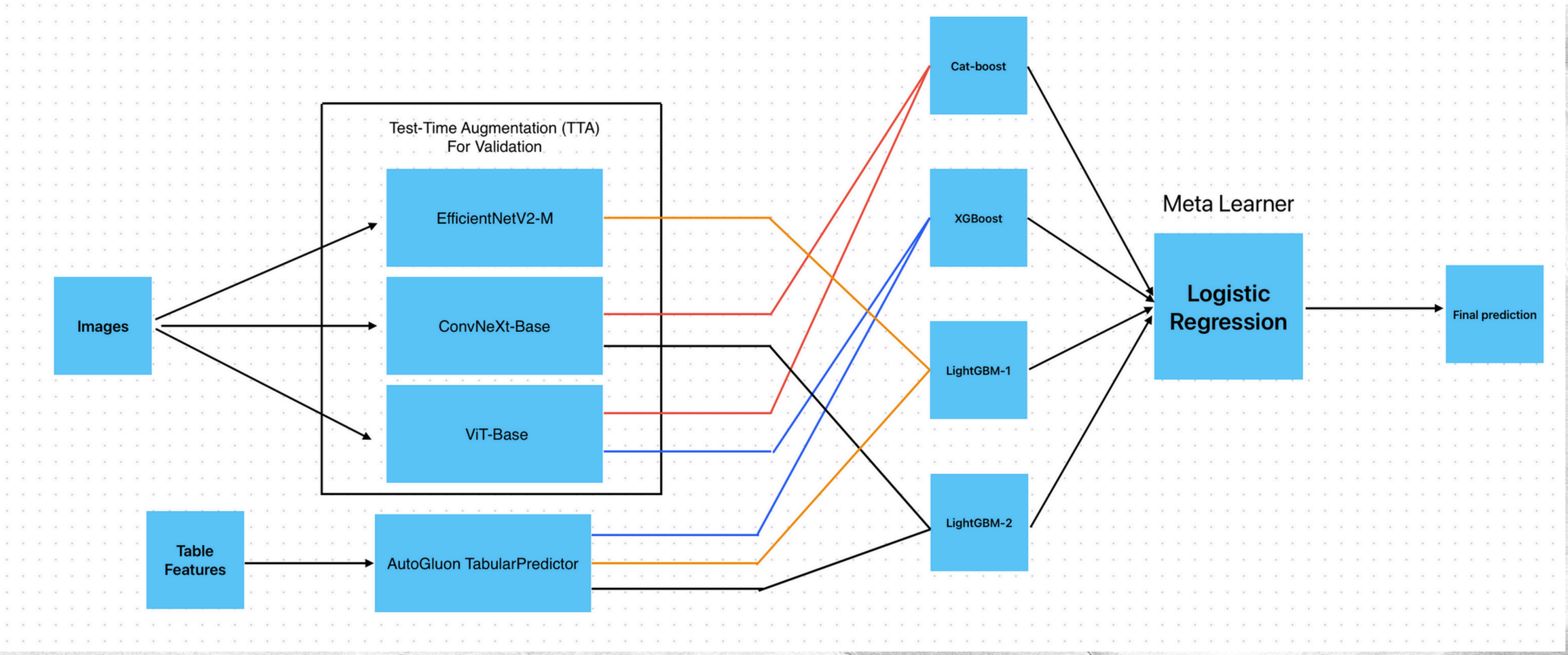


Image Backbone Selection

EfficientNetV2-M, ConvNeXt-Base, and ViT-Base were the three models chosen ,

They were chosen to maximise diversity

Effnet

Compound scaling of depth, width, and resolution for balanced efficiency

Systematically balances network depth (number of layers), width channels per layer), and input image resolution to maximize accuracy per FLOP.

Mobile inverted bottleneck (MBConv)

An MBConv block starts by expanding a narrow input (few channels) to a higher-dimensional space, applies a depthwise separable convolution (one convolution per channel) to capture spatial features efficiently, and then projects back to a narrow output to reduce computation while maintaining representational power.

ConvNext

ResNet-style convolutional backbone modernized with Transformer-inspired design choices

Expands channels then projects back down ,increasing representational capacity without excessive compute.

Large convolutional kernels (7×7) and inverted bottlenecks

Captures wider spatial context per layer, similar to self-attention’s receptive field, enhancing feature aggregation over larger image regions

Layer normalization replacing batch norm

Expands channels then projects back down (opposite of classic ResNet), increasing representational capacity without excessive compute.

ViT

Vision Transformer using self-attention over fixed-size image patches

Divides the image into patches and applies attention to compute pairwise patch relationships, enabling global context modeling from the outset.

Global context modeling across entire image

Each token attends to every other token at every layer, capturing long-range dependencies and holistic scene understanding.

Multi-head attention layers for capturing long-range dependencies

Employs multiple parallel attention “heads” to learn diverse feature subspaces simultaneously, improving the model’s ability to capture varied patterns

Diverse Feature Representation

Convolutions specialize in capturing local patterns such as edges, textures, and fine-grained gradients.
Transformers excel at modeling non-local interactions, capturing global spatial relationships, and forming higher-order semantic groupings

Pretrained models are available, Hence optimized implementations are readily available, enabling fine tuning on them was simply sufficient for good performance

Test-Time Augmentation (TTA)

Each image is transformed into multiple augmented variants at the time of validation

This is done here to reduce the sensitivity to input noise

Also it helps in decreasing overfitting

AutoGluon Tabular feature Engineering

Categorical Encoding

Converts nominal and ordinal variables via one-hot, target, and frequency encodings,

Feature Crossing

makes interaction features that capture non linear relations

Imputation and Outlier Handling

imputes missing values using model-based or statistical methods and caps extreme outliers to reduce noise

Binning and Scaling

Discretizes continuous variables into quantiles or domain-informed buckets, normalizes skewed distributions, and applies transformations like log where beneficial.

Overfitting Control

It also prevents overfitting while generation of large feature sets by Regularized Base Models,Feature Pruning I.e. low importance to features with less information gain and also using cross validation and ensembling

Base Learner Model

CatBoost

Input: ConvNeXt Features + ViT Features

It is only given Image features and not given any tabular features ,Hence enabling it to predict the most deep image patterns

CatBoost is specifically chosen for this job as

The two models ConvNext and Vit ,even though being architecturally different the conv model builds on going from local feature to global context whereas transformer builds on the global context from the start ,even though there would still be many overlapping features hence making the data to be Noisy

cat boost is the perfect model for this noisy data as it uses oblivious tree ,which builds symmetric trees where regularisation is highly prioritised

XgBoost

Input:ViT Features + AutoGluon Tabular Features

The Transformer model gives out highly complex features which are to be added with the autogluon tabular features

XGBoost's Architecture of level wise tree growth makes it possible to train deeper trees with included regularisation.Hence enabling formation of. highly interactive features making

it the best suitable model for this combination.

LightGBM-1 & 2

Input: Effnet/ConvNext features + AutoGluon Tabular Features

Both Effnet and ConvNext are CNNs which have a hierarchical order of detecting features in an image starting from local features like edges, textures to larger patterns, these are to be added to Autogluon tabular features.

LGBM is a model that uses Best first tree growth approach where at any tree the node that gives high impact i.e. largest information gain is split first

This would not go well with the Transformer architecture as the layers there are not ordered hence making it unnecessarily too complex ,whereas due to an ordered nature of convolutions LGBM's efficient architecture could be utilised.

Also OPTUNA was used for 30 trials for automated hyperparameter optimisation in the GBDTs

Meta-Learner

Here a simple model like Logistic Regression is used instead of a complex model like GBDT’s as
in a stacking problem the problem of overfitting is prominent ,Hence using a simpler linear model that is not overly flexible does the job right

Here the Logistic Regression model acts as a weighted average for the four base learners

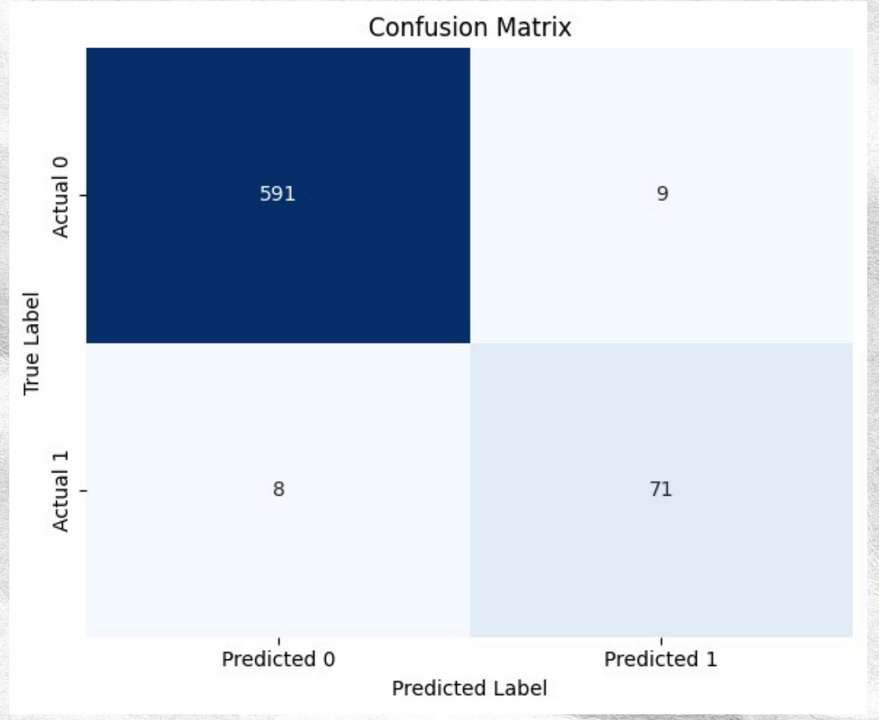
Hold out test set predictions

```
--- Final Hold-Out Test Set Evaluation Summary ---
Accuracy:  0.9750
Precision: 0.8875
Recall:    0.8987
F1-score:  0.8931
ROC AUC:   0.9935

Hold-Out Test Set Classification Report:
              precision    recall  f1-score   support

           0       0.99      0.98      0.99       600
           1       0.89      0.90      0.89        79

    accuracy                           0.97       679
   macro avg       0.94      0.94      0.94       679
weighted avg       0.98      0.97      0.98       679
```



THANKYOU